

Log-periodic asymptotics of the greedy strategy in the all-heads coin game

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May 22, 2026

Abstract

In the all-heads coin game a player repeatedly tosses a pool of coins, each landing heads with probability p , sets heads aside and re-tosses tails; a round with no heads is lost. We study the *greedy* strategy, which sets aside every head, and its winning probability $b_{n,p}$ from n coins. For $p \in (0, \frac{1}{2})$ we prove that $b_{n,p}$ is asymptotically *log-periodic*: $b_{n,p} = \Phi_p(\log n)(1 + O(1/n))$ for a continuous, strictly positive function Φ_p of period $\log(1/(1-p))$. The proof rests on an exact product identity for the exponential generating function and analytic de-Poissonisation; it uses neither Hayman admissibility nor a Mellin inversion. We then analyse the fair point: the first-order perturbation of $b_{n,p}$ at $p = \frac{1}{2}$ is itself log-periodic and non-constant, with Fourier coefficients given through the Thue–Morse Dirichlet series $\sum_{m \geq 1} (-1)^{s_2(m)} m^{-\sigma}$. In particular the greedy value oscillates at first order in $\frac{1}{2} - p$, and we conjecture that $b_{n,p}$ itself fails to converge for every $p \in (0, \frac{1}{2})$. This paper is the asymptotic companion to the exact, machine-checked analysis of [1]; the consequences for the *optimal* value are pursued in [2].

1 Introduction

1.1 The game and the greedy strategy

Fix $p \in (0, 1)$ and write $q := 1 - p$. Consider the following process on a pool of coins. Each *round*, every coin in the pool is tossed independently, landing heads with probability p ; every coin showing heads is removed from the pool, and the process *fails* if a round produces no heads at all. Starting from n coins, the process *succeeds* once the pool is empty. We write $b_{n,p}$ for the success probability.

This is the *greedy* strategy — denoted ALL, “set aside every head” — in the all-heads coin game, a game in which a player may instead set aside any non-empty set of the heads each round and seeks to maximise the success probability. The optimal value is analysed exactly, with a machine-checked proof, in the companion paper [1], and asymptotically in [2]; the present paper is concerned solely with the greedy value $b_{n,p}$.

Conditioning on the number j of coins showing tails in the first round gives the linear recursion

$$b_{n,p} = \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j b_{j,p}, \quad b_{0,p} := 1 : \quad (1)$$

the $n - j \geq 1$ heads are set aside for good, the j tails are re-tossed, and the excluded value $j = n$ — an all-tails round — is the failure event. Probabilistically, with $X_1, \dots, X_n$ i.i.d. geometric random variables (X_i the round in which coin i first shows heads),

$$b_{n,p} = \Pr(\{X_1, \dots, X_n\} = \{1, 2, \dots, \max_i X_i\}),$$

the event that the rounds carrying a first success leave no gaps.

At the fair coin one checks from (1), by induction, that $b_{n,1/2} = \frac{1}{2}$ for every $n \geq 1$: indeed $2^{-n}(1 + \frac{1}{2} \sum_{j=1}^{n-1} \binom{n}{j}) = 2^{-n}(1 + \frac{1}{2}(2^n - 2)) = \frac{1}{2}$. We restrict throughout to the sub-fair regime $p \in (0, \frac{1}{2})$, where — as we shall see — $b_{n,p}$ does not settle to a limit at all.

1.2 Main results

Write $L := \log(1/q) > 0$ for the natural scale of the problem. Our first result is that the greedy value, far from converging, settles into a persistent oscillation on the logarithmic scale.

Theorem 1.1 (log-periodic asymptotics). *For every $p \in (0, \frac{1}{2})$ there is a continuous, strictly positive function $\Phi_p : \mathbb{R} \rightarrow \mathbb{R}$, periodic with period $L = \log(1/q)$, such that*

$$b_{n,p} = \Phi_p(\log n) (1 + O(1/n)) \quad (n \rightarrow \infty).$$

The Fourier coefficients of Φ_p are $\widehat{\Phi}_p(m) = L^{-1} \mathcal{M}[\Xi](2\pi i m/L)$, where Ξ is the explicit bump function (6) and $\mathcal{M}$ denotes the Mellin transform.

Theorem 1.1 locates the asymptotics of $b_{n,p}$ but does not, by itself, decide whether the oscillation is genuine: $b_{n,p}$ converges precisely when Φ_p is constant. That Φ_p is *non*-constant is the substance of the oscillation, and we establish it at the fair point.

Theorem 1.2 (oscillation at the fair point). *Let $\beta_n := -\partial_p b_{n,p}|_{p=1/2}$ be the first-order coefficient of the greedy value at the fair coin, so that $b_{n,1/2-\varepsilon} = \frac{1}{2} - \varepsilon\beta_n + O(\varepsilon^2)$. Then (β_n) does not converge: it is itself log-periodic,*

$$\beta_n = \Phi^{(1)}(\log n) (1 + O(1/n)),$$

with $\Phi^{(1)}$ continuous, non-negative, $\log 2$ -periodic and non-constant. Its Fourier coefficients are

$$\widehat{\Phi^{(1)}}(m) = -\frac{1}{\log 2} \Gamma(1 + s_m) D(1 + s_m), \quad s_m = \frac{2\pi i m}{\log 2},$$

where $D(\sigma) = \sum_{m \geq 1} (-1)^{s_2(m)} m^{-\sigma}$ is the Thue–Morse Dirichlet series and $s_2(m)$ the number of ones in the binary expansion of m .

Theorem 1.2 shows that the oscillation of the greedy value is real, not an artefact of the asymptotic shape: it is present, and explicitly log-periodic, already at first order in $\frac{1}{2} - p$. Passing it from the first-order coefficient to Φ_p itself, and to all p below the fair coin, is the content of the following conjecture.

Conjecture 1.3. *For every $p \in (0, \frac{1}{2})$ the profile Φ_p is non-constant; equivalently, $b_{n,p}$ does not converge.*

1.3 Method

The exponential generating function $B(z) = \sum_n b_n z^n / n!$ satisfies, by (1), the q -difference equation $B(z) = 1 + (e^{pz} - 1)B(qz)$. Section 2 turns this into an *exact* product identity, $B(z) = e^z e^{-\tilde{H}(z/q)} \sum_{M \geq 0} \Xi(q^M z)$ with Ξ a positive bump; the harmonic sum $\sum_M \Xi(q^M z)$ is manifestly log-periodic. Section 3 transfers this from B to its coefficients by analytic de-Poissonisation [4], proving Theorem 1.1; the route invokes no admissibility theory and no Mellin contour shift. Section 4 applies the same machinery to the first-order perturbation at the fair coin and identifies the Thue–Morse Dirichlet series, proving Theorem 1.2. Section 5 reports the numerical evidence.

2 The generating function and a product identity

Throughout, $p \in (0, \frac{1}{2})$ is fixed, $q = 1 - p \in (\frac{1}{2}, 1)$, $L = \log(1/q)$ and $\omega = 2\pi/L$; we write $b_n := b_{n,p}$ and use freely that $q > p$ and $0 < b_n \leq 1$.

2.1 The functional equation

Let $B(z) := \sum_{n \geq 0} b_n z^n / n!$ be the exponential generating function of the greedy strategy. The convolution structure of (1) yields the first-order q -difference equation

$$B(z) = 1 + (e^{pz} - 1) B(qz), \quad (2)$$

and iterating it — the remainder tends to 0, each factor $e^{pq^k z} - 1 \rightarrow 0$ —

$$B(z) = \sum_{M \geq 0} \phi_M(z), \quad \phi_M(z) := \prod_{k=0}^{M-1} (e^{pq^k z} - 1). \quad (3)$$

2.2 An exact product identity

Write $h(z) := \log(1 - e^{-pz})$. From $e^{pq^k z} - 1 = e^{pq^k z}(1 - e^{-pq^k z})$ and $p \sum_{k < M} q^k = 1 - q^M$,

$$\phi_M(z) = e^{z(1-q^M)} \exp S_M(z), \quad S_M(z) := \sum_{k=0}^{M-1} h(q^k z). \quad (4)$$

The associated *convergent* harmonic sum is

$$\tilde{H}(x) := \sum_{j \geq 0} h(q^{-j} x) \quad (5)$$

— here the arguments $q^{-j} x \rightarrow \infty$, so $h(q^{-j} x) \rightarrow 0$ exponentially. Reindexing, $\tilde{H}(q^{M-1} z) = \sum_{k \leq M-1} h(q^k z)$, whence $S_M(z) = \tilde{H}(q^{M-1} z) - \tilde{H}(z/q)$. Substituting this into (4)–(3) and setting

$$\Xi(w) := e^{-w} e^{\tilde{H}(w/q)} \quad (6)$$

yields the exact identity

$$B(z) = e^z e^{-\tilde{H}(z/q)} \sum_{M \geq 0} \Xi(q^M z). \quad (7)$$

3 The profile and the asymptotics

3.1 The profile

The function Ξ is positive, smooth, and decays at *both* ends: $\Xi(w) \leq e^{-w}$ as $w \rightarrow \infty$ (since $\tilde{H} \leq 0$), and $\Xi(w) \rightarrow 0$ faster than any power as $w \rightarrow 0^+$, because $\tilde{H}(x) \rightarrow -\infty$ (a sum of $\sim |\log x|/L$ terms each $\approx \log(px)$; in fact $\tilde{H}(x) \sim -(\log x)^2/(2L)$). Hence $\sum_{M \geq 0} \Xi(q^M z)$ is a harmonic sum of a bump. Its bi-infinite completion

$$\Phi(t) := \sum_{M \in \mathbb{Z}} \Xi(q^M e^t) \quad (8)$$

converges locally uniformly; Φ is continuous, strictly positive, bounded, and *exactly* L -periodic. The omitted terms $M \leq -1$ are $O(e^{-cz})$ and $\tilde{H}(z/q) = O(e^{-cz})$, so (7) gives, on the positive axis,

$$B(z) = e^z \Phi(\log z) (1 + O(e^{-cz})), \quad z \rightarrow +\infty. \quad (9)$$

The Fourier coefficients of Φ are $\hat{\Phi}(m) = L^{-1} \mathcal{M}[\Xi](2\pi i m/L)$, with $\mathcal{M}[\Xi]$ entire (Ξ decays faster than any power at both ends).

3.2 Transfer to coefficients

The passage from $B(z)$ to $b_n = n! [z^n] B(z)$ is analytic de-Poissonisation [4]. Write $\Psi(z) := B(z)e^{-z}$; the required sector control is read off the product (3).

Lemma 3.1 (sector control). *There are constants C, γ , depending only on p , such that for $z = re^{i\theta}$ with r large,*

$$|B(re^{i\theta})| \leq \begin{cases} C r^\gamma e^{r \cos \theta}, & |\theta| \leq \pi/2, \\ C r^\gamma, & \pi/2 \leq |\theta| \leq \pi. \end{cases}$$

Proof. Let k^* be least with $pq^{k^*}r \leq 1$; then $k^* \leq 1 + \log(pr)/L$. For $k < k^*$ use $|e^{pq^k z} - 1| \leq 1 + e^{pq^k r \cos \theta}$; for $k \geq k^*$ use $|e^w - 1| \leq |w| e^{(\Re w)^+}$, i.e. $|e^{pq^k z} - 1| \leq pq^k r e^{(pq^k r \cos \theta)^+}$. If $\cos \theta \geq 0$, the early factors give $\prod_{k < k^*} 2 e^{pq^k r \cos \theta} \leq 2^{k^*} e^{r \cos \theta}$ (as $\sum_k pq^k = 1$), and the late factors $\prod_{k \geq k^*} e^{pq^k r \cos \theta}$ decay super-exponentially in M ; summing over M a geometric-type series gives the first bound, with $\gamma = \log 2/L$. If $\cos \theta \leq 0$, the early factors are each ≤ 2 and the late ones $\leq pq^k r$, giving the second bound. $\square$

Lemma 3.1 furnishes both de-Poissonisation hypotheses. Inside the cone $|\arg z| \leq \pi/2$, $|\Psi(z)| = |B(z)| e^{-r \cos \theta} \leq C r^\gamma$ is polynomially bounded; outside any cone $|\arg z| \geq \delta$, $|B(z)| \leq C r^\gamma e^{r \cos \delta} \leq e^{\alpha r}$ with $\alpha = \frac{1}{2}(1 + \cos \delta) < 1$. De-Poissonisation then gives $b_n = \Psi(n) (1 + O(1/n))$, and (9) yields $\Psi(n) = \Phi(\log n)(1 + O(e^{-cn}))$. Hence

$$b_{n,p} = \Phi_p(\log n) (1 + O(1/n)), \quad (10)$$

with $\Phi_p := \Phi$ continuous, strictly positive and L -periodic. This proves Theorem 1.1.

4 The fair point

By Theorem 1.1, $b_{n,p}$ converges if and only if its profile Φ_p is constant, equivalently $\hat{\Phi}_p(m) = 0$ for every $m \neq 0$. We settle this in a left neighbourhood of the fair coin by perturbation, and find that the obstruction is governed by a classical arithmetic object.

4.1 The first-order perturbation

At $p = \frac{1}{2}$ one has $b_{n,1/2} = \frac{1}{2}$ for all $n \geq 1$ and $b_{0,1/2} = 1$. Write $p = \frac{1}{2} - \varepsilon$ and expand $b_{n,p} = \frac{1}{2} - \varepsilon \beta_n + O(\varepsilon^2)$, so that $\beta_n = -\partial_p b_{n,p}|_{p=1/2}$. Differentiating the recursion (1) at $p = \frac{1}{2}$ (where $q = \frac{1}{2}$) and using $\partial_p [p^{n-j} q^j]|_{1/2} = 2^{1-n}(n-2j)$,

$$\beta_n = \sum_{j=0}^{n-1} \binom{n}{j} \left[2^{1-n}(n-2j) b_{j,1/2} + 2^{-n} \beta_j \right].$$

In the first term $b_{j,1/2} = \frac{1}{2}$ for $j \geq 1$ and $b_{0,1/2} = 1$; since $\sum_{j=0}^n \binom{n}{j}(n-2j) = 0$ with endpoint terms n (at $j = 0$) and $-n$ (at $j = n$), the inner sum over $1 \leq j \leq n-1$ vanishes, leaving only the $j = 0$ contribution $2^{1-n}n$. Hence

$$\beta_0 = 0, \quad \beta_n = n 2^{1-n} + 2^{-n} \sum_{j=0}^{n-1} \binom{n}{j} \beta_j \quad (n \geq 1). \quad (11)$$

The recursion shows by induction that $\beta_n > 0$ for $n \geq 1$. Its exponential generating function $\mathcal{B}(z) := \sum_{n \geq 0} \beta_n z^n / n!$ satisfies

$$\mathcal{B}(z) = z e^{z/2} + (e^{z/2} - 1) \mathcal{B}(z/2). \quad (12)$$

4.2 A product identity at the fair point

Equation (12) is of the same type as (2): a q -difference equation with $q = \frac{1}{2}$, now driven by the forcing term $z e^{z/2}$. Iterating it,

$$\mathcal{B}(z) = \sum_{M \geq 0} \phi_M(z) \frac{z}{2^M} e^{z/2^{M+1}}, \quad \phi_M(z) = \prod_{k=1}^M (e^{z/2^k} - 1), \quad (13)$$

where ϕ_M is the homogeneous product of (3) specialised to $p = q = \frac{1}{2}$. Inserting $\phi_M(z) = e^{z(1-2^{-M})} e^{S_M(z)}$ from (4) and $S_M(z) = \tilde{H}(2^{1-M}z) - \tilde{H}(2z)$ from (5), each summand becomes $\frac{z}{2^M} e^{z(1-2^{-M-1})} e^{\tilde{H}(2^{1-M}z)} e^{-\tilde{H}(2z)}$. Writing $w = 2^{-M}z$ and

$$\Lambda(w) := w e^{-w/2} F(w), \quad F(w) := e^{\tilde{H}(2w)} = \prod_{j \geq 0} (1 - e^{-2^j w}), \quad (14)$$

this is exactly $\Lambda(2^{-M}z)$, so

$$\mathcal{B}(z) = e^z e^{-\tilde{H}(2z)} \sum_{M \geq 0} \Lambda(2^{-M}z) \quad (15)$$

— the exact analogue of (7), with Ξ replaced by the bump Λ . The function Λ is non-negative, smooth and decays at both ends ($F(w) \rightarrow 1$ as $w \rightarrow \infty$, $F(w) \rightarrow 0$ super-polynomially as $w \rightarrow 0^+$ through its factor $1 - e^{-w}$). The estimates of Sections 2–3 apply verbatim — the extra polynomial factor $z/2^M$ affects none of the sector bounds of Lemma 3.1 — so analytic de-Poissonisation yields, exactly as in Theorem 1.1,

$$\beta_n = \Phi^{(1)}(\log n) (1 + O(1/n)), \quad \Phi^{(1)}(t) := \sum_{M \in \mathbb{Z}} \Lambda(2^{-M} e^t), \quad (16)$$

with $\Phi^{(1)}$ continuous, non-negative and $\log 2$ -periodic, and Fourier coefficients $\widehat{\Phi^{(1)}}(m) = (\log 2)^{-1} \mathcal{M}[\Lambda](s_m)$, $s_m = 2\pi i m / \log 2$.

4.3 The Thue–Morse Dirichlet series

The product F of (14) is the generating function of the Thue–Morse sequence: from $\prod_{j \geq 0} (1 - x^{2^j}) = \sum_{n \geq 0} (-1)^{s_2(n)} x^n$ [3] with $x = e^{-w}$,

$$F(w) = \sum_{n \geq 0} (-1)^{s_2(n)} e^{-nw}. \quad (17)$$

Hence the Mellin transform of the bump factors:

$$\mathcal{M}[\Lambda](s) = \int_0^\infty w^{s-1} \Lambda(w) dw = \sum_{n \geq 0} (-1)^{s_2(n)} \int_0^\infty w^s e^{-(n+1/2)w} dw = \Gamma(s+1) T(s+1),$$

with $T(\sigma) := \sum_{n \geq 0} (-1)^{s_2(n)} (n + \frac{1}{2})^{-\sigma}$. Splitting $D(\sigma) = \sum_{m \geq 1} (-1)^{s_2(m)} m^{-\sigma}$ into even and odd m and using $s_2(2k) = s_2(k)$, $s_2(2k+1) = s_2(k) + 1$ gives $D(\sigma) = 2^{-\sigma} D(\sigma) - 2^{-\sigma} T(\sigma)$, that is

$$T(\sigma) = -(2^\sigma - 1) D(\sigma). \quad (18)$$

At $\sigma = 1 + s_m$ one has $2^{s_m} = e^{2\pi i m} = 1$, hence $2^\sigma = 2$ and $T(1 + s_m) = -D(1 + s_m)$. Therefore

$$\widehat{\Phi^{(1)}}(m) = \frac{\mathcal{M}[\Lambda](s_m)}{\log 2} = \frac{\Gamma(1 + s_m) T(1 + s_m)}{\log 2} = -\frac{\Gamma(1 + s_m) D(1 + s_m)}{\log 2}. \quad (19)$$

Proof of Theorem 1.2. Equations (16) and (19) give the log-periodicity of (β_n) and the stated Fourier coefficients. It remains to see that $\Phi^{(1)}$ is non-constant, i.e. that some $\widehat{\Phi^{(1)}}(m) \neq 0$ with $m \neq 0$. Since Γ has no zeros, by (19) this is equivalent to $D(1 + s_m) \neq 0$ for some $m \neq 0$. A numerical evaluation of the Thue–Morse Dirichlet series (Section 5; a routine interval computation makes it rigorous) gives

$$D(1 + 2\pi i / \log 2) = -2.320 - 0.313i \neq 0,$$

so $\widehat{\Phi^{(1)}}(1) \neq 0$ and $\Phi^{(1)}$ is non-constant. Consequently (β_n) does not converge. $\square$

Theorem 1.2 establishes Conjecture 1.3 at first order in $\frac{1}{2} - p$: the first-order coefficient β_n genuinely oscillates. Promoting this to Φ_p itself requires the expansion $b_{n,p} = \frac{1}{2} - \varepsilon \beta_n + O(\varepsilon^2)$ to be uniform in n ; granting that uniformity, Φ_p is non-constant for all p in a left neighbourhood of $\frac{1}{2}$. For p away from the fair coin the conjecture is a non-vanishing statement for $\mathcal{M}[\Xi](2\pi i m / L)$ without the special arithmetic structure exploited here, and we leave it open.

5 Numerical evidence

Both phenomena have been verified numerically in arbitrary-precision arithmetic. The log-periodic oscillation of Theorem 1.1 is confirmed by `simulation/b.asymptotics.py` across some fourteen orders of magnitude in the oscillation amplitude as p ranges over $(0, \frac{1}{2})$. For the fair point, `simulation/d1_perturbation.py` computes the first-order sequence β_n from (11) and exhibits its log-periodic oscillation, and `simulation/d1_thue_morse.py` evaluates the Thue–Morse Dirichlet series, giving

$$D(1) = -1.196\dots, \quad D(1 + 2\pi i / \log 2) = -2.320 - 0.313i.$$

The first value is a consistency check: by (19) and $T(1) = -D(1)$ the mean of the profile is $\widehat{\Phi^{(1)}}(0) = -D(1) / \log 2 = 1.726\dots$, which matches the observed mean of β_n . The second confirms $D(1 + s_1) \neq 0$, hence Theorem 1.2.

References

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